

ENDG 319

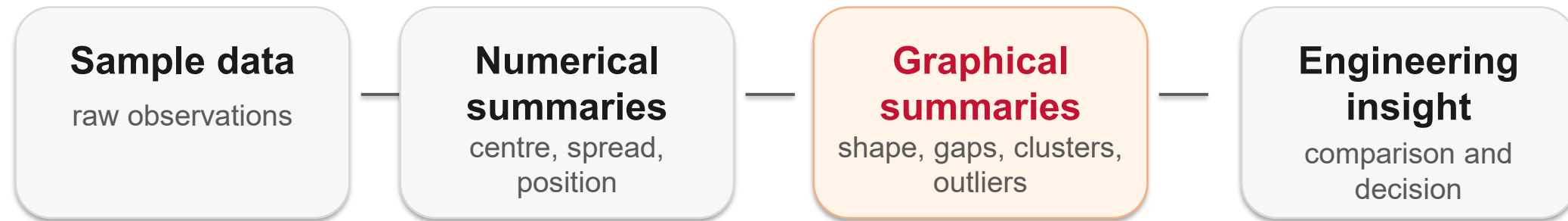
Probability, Statistics and Machine Learning

Lecture 3: Graphical Summaries of Data

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From numerical summaries to graphical summaries

A few numbers summarize a dataset—but a graph reveals its structure.



Use both numerical and graphical summaries: each answers questions the other may miss.

Learning outcomes for Lecture 3

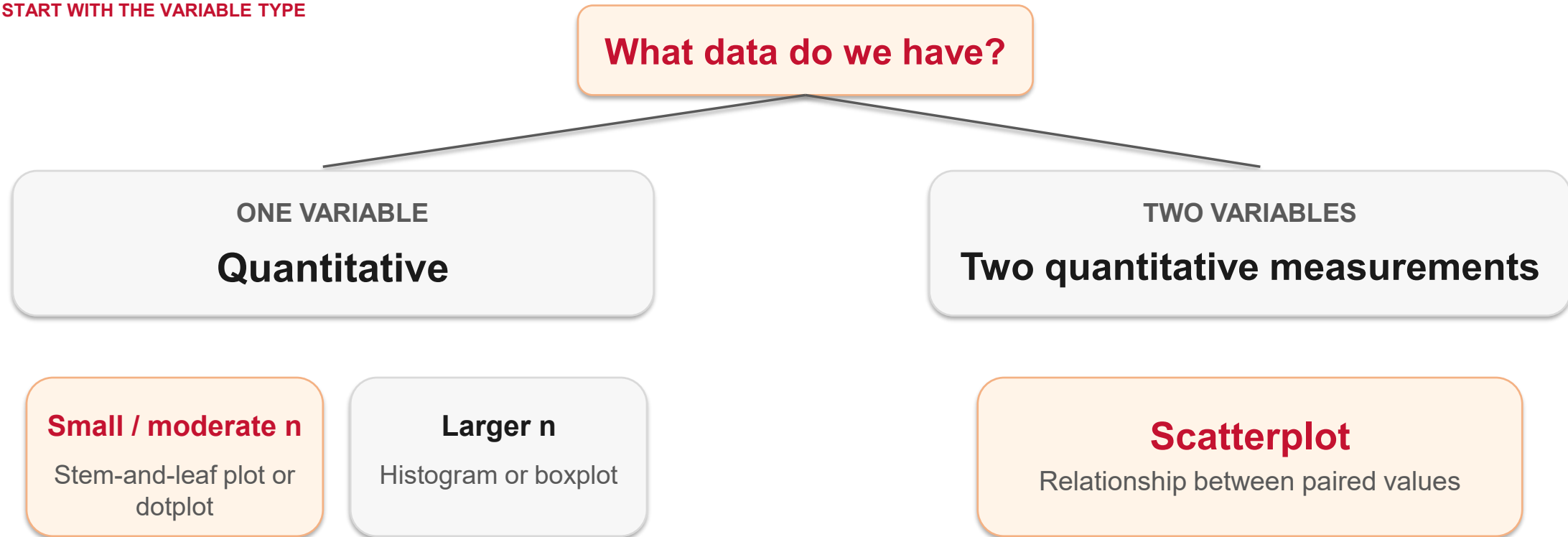
BY THE END OF TODAY, STUDENTS SHOULD BE ABLE TO

- Construct and interpret frequency tables for quantitative data.
- Create and read stem-and-leaf plots, dotplots, histograms, and boxplots.
- Describe distribution shape using symmetry, skewness, and modality.
- Use comparative boxplots to compare centre, spread, shape, and outliers.
- Use scatterplots to examine relationships between two quantitative variables.

Core skill: choose a graphical summary that matches the data and the engineering question.

Choose the display to match the data

START WITH THE VARIABLE TYPE



Categorical data are summarized with frequency tables and bar charts—not histograms.

What should a graphical summary reveal?

CENTRE

Where are values concentrated?

SPREAD

How variable are the observations?

SHAPE

Is the distribution symmetric, skewed, or multimodal?

UNUSUAL FEATURES

Are there gaps, clusters, or outliers?

RELATIONSHIP

Do two variables move together?

A good graph supports an interpretation; it does not replace one.

Frequency tables for continuous data

When a sample is large, group observations into non-overlapping class intervals.

CLASS FREQUENCY

Number of sample values in a class interval

$$f_i$$

RELATIVE FREQUENCY

Proportion of the sample in the class

$$f_i / n$$

DENSITY

Relative frequency per unit of class width

$$(f_i / n) / w_i$$

Class width w_i = upper class limit – lower class limit

For equal-width classes, relative frequency and density differ only by the common width.

Example 6: undesired side-product yield

Twenty-four reaction runs produced the following undesired side-product yields (%):

4.4	3.4	2.6	3.8	4.9	4.6
5.2	4.7	4.1	2.6	6.7	4.1
3.6	2.9	2.6	4.0	4.3	3.9
4.8	4.5	4.4	3.1	5.7	4.5

MINIMUM

2.6

MAXIMUM

6.7

SAMPLE SIZE

$n = 24$

GOAL

Summarize shape

First question: how should we group the values into class intervals?

Example 6: choose class intervals

One practical choice is five equal-width classes.

NUMBER OF CLASSES

$$\sqrt{n} = \sqrt{24} \approx 5$$

A heuristic—not a universal rule

CLASS WIDTH

$$(7 - 2) / 5 = 1$$

Convenient limits: 2 to 7

CLASS INTERVALS

$$2 \leq x < 3$$

$$3 \leq x < 4$$

$$4 \leq x < 5$$

$$5 \leq x < 6$$

$$6 \leq x < 7$$

All observations must fall in exactly one interval.

Changing the class limits or number of classes can change the visual impression.

Example 6: frequency, relative frequency, and density

Class (% yield)	Midpoint	Frequency	Relative frequency	Density
$2 \leq x < 3$	2.5	4	0.17	0.17
$3 \leq x < 4$	3.5	5	0.21	0.21
$4 \leq x < 5$	4.5	12	0.50	0.50
$5 \leq x < 6$	5.5	2	0.08	0.08
$6 \leq x < 7$	6.5	1	0.04	0.04

INTERPRET THE TABLE

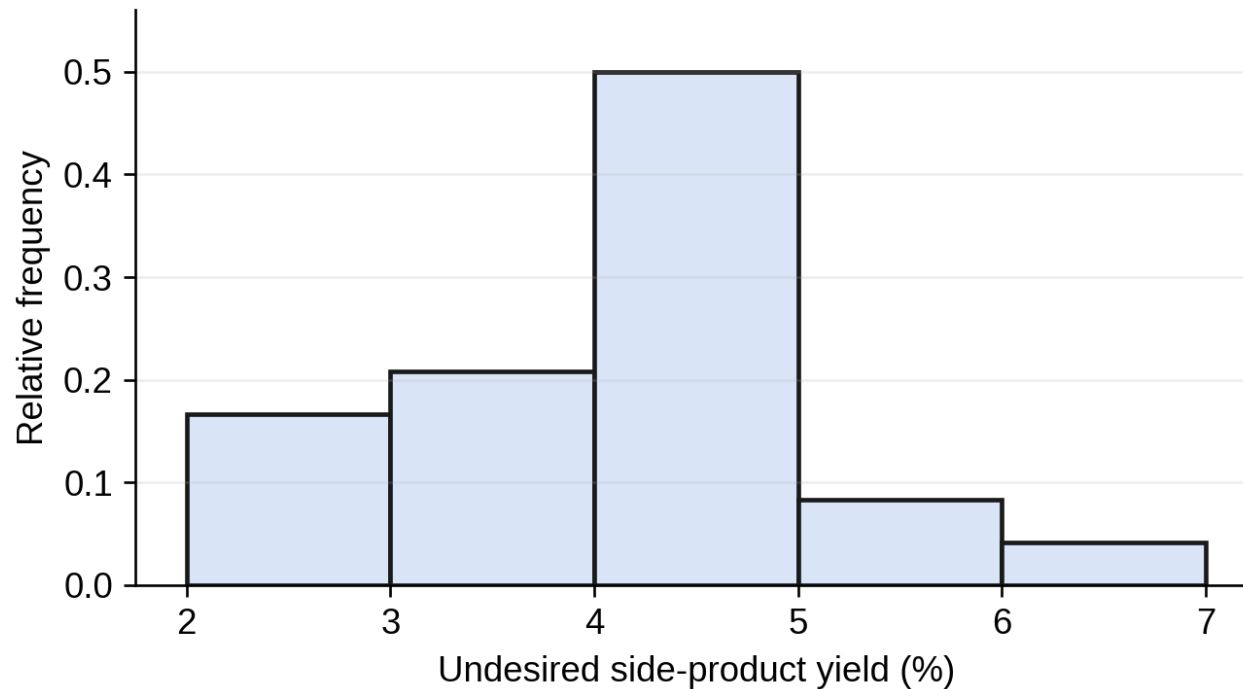
What proportion of runs had yield at least 5%?

$$(2 + 1) / 24 \\ = 0.125 = 12.5\%$$

Checks: frequencies sum to 24; relative frequencies sum to approximately 1.

From a frequency table to a histogram

Each class interval becomes a bar; adjacent bars touch because the measurement scale is continuous.

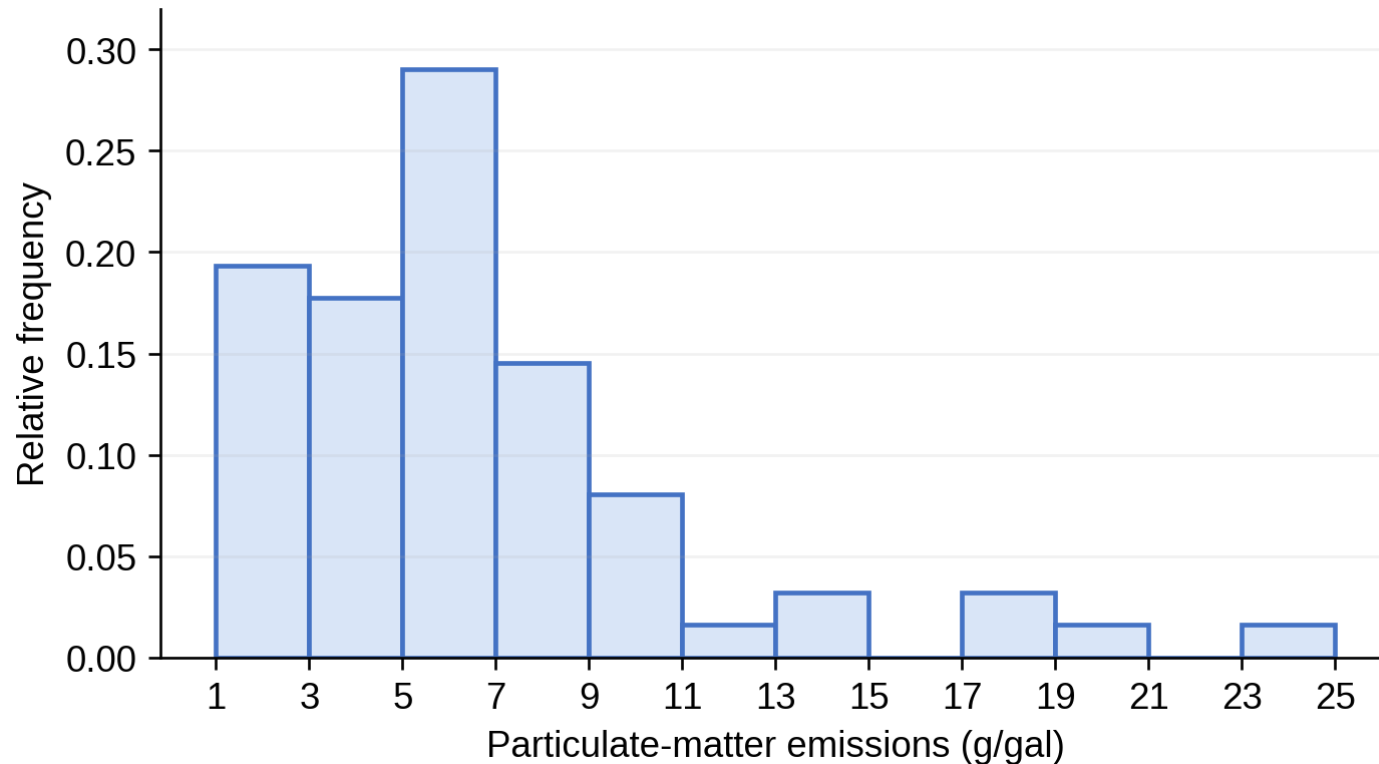


READ THE GRAPH

- Peak** 4%—<5%
- Half the runs** fall in that class
- Right tail** extends to 6.7%
- At least 5%** 3 of 24 runs

The histogram makes concentration and tail behaviour visible immediately.

Histogram anatomy: classes, peaks, and tails



CLASS INTERVAL

Example: $1 \leq x < 3$

PEAK

Region of greatest concentration

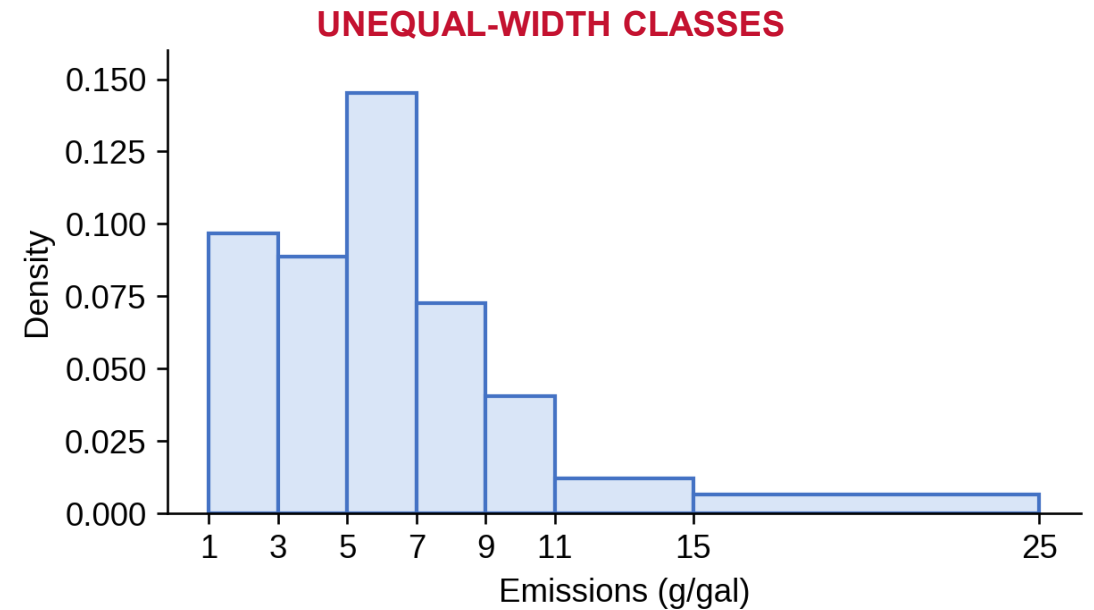
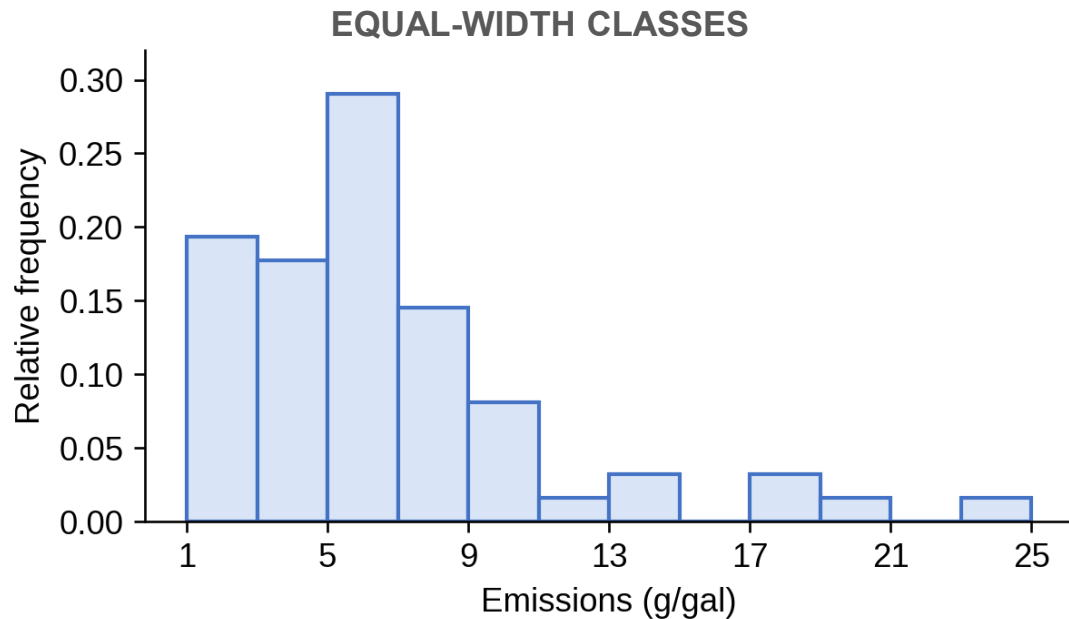
TAIL

Sparse values extending from the centre

Bar height alone represents frequency only when the class widths are equal.

Unequal-width histograms: area carries the frequency

When class widths differ, bar area—not bar height—must equal relative frequency.

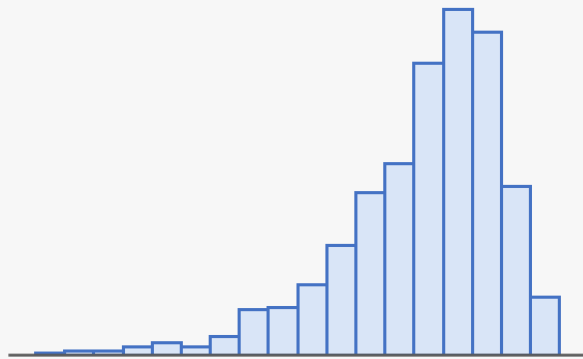


$$\text{Area} = \text{width} \times \text{density} = \text{relative frequency}$$

Symmetry and skewness

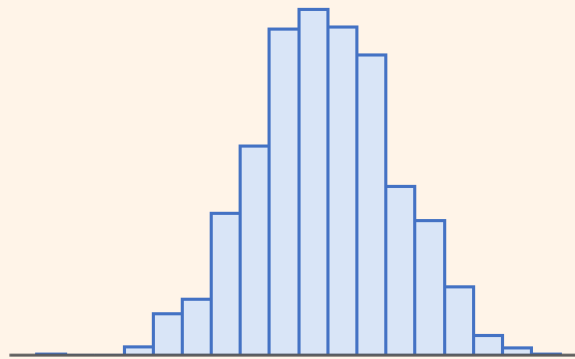
Skewness is named for the direction of the longer tail.

LEFT-SKEWED



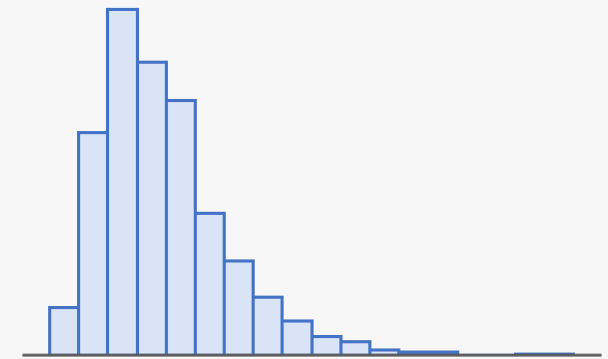
long tail to the left

SYMMETRIC



halves are approximately mirror
images

RIGHT-SKEWED

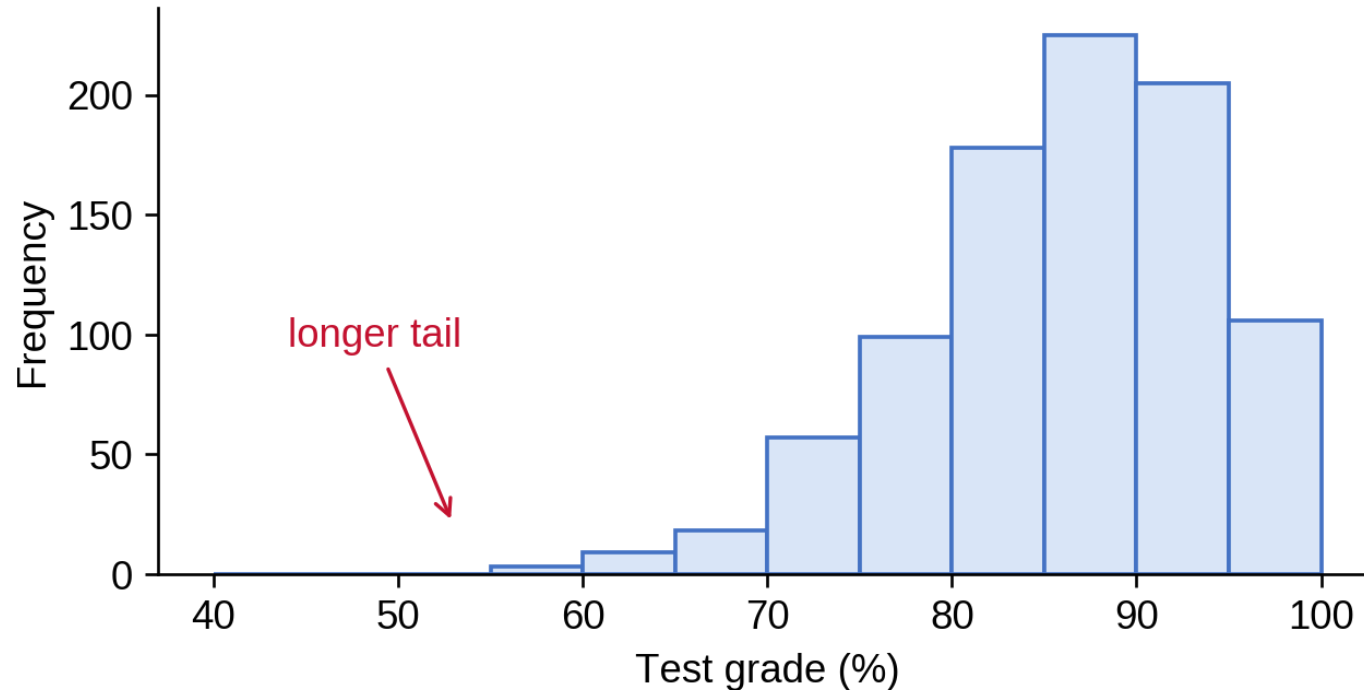


long tail to the right

Do not name skewness by where most observations lie; name it by the tail.

Concept check: grades on an easy test

Most students score high; a smaller number receive much lower grades.



QUESTION

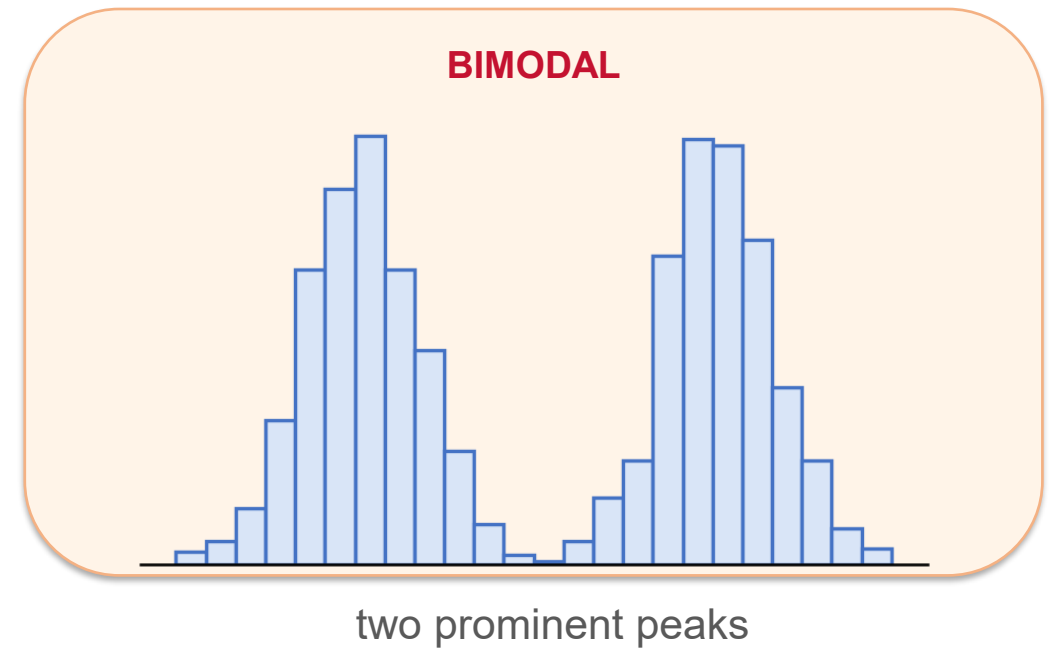
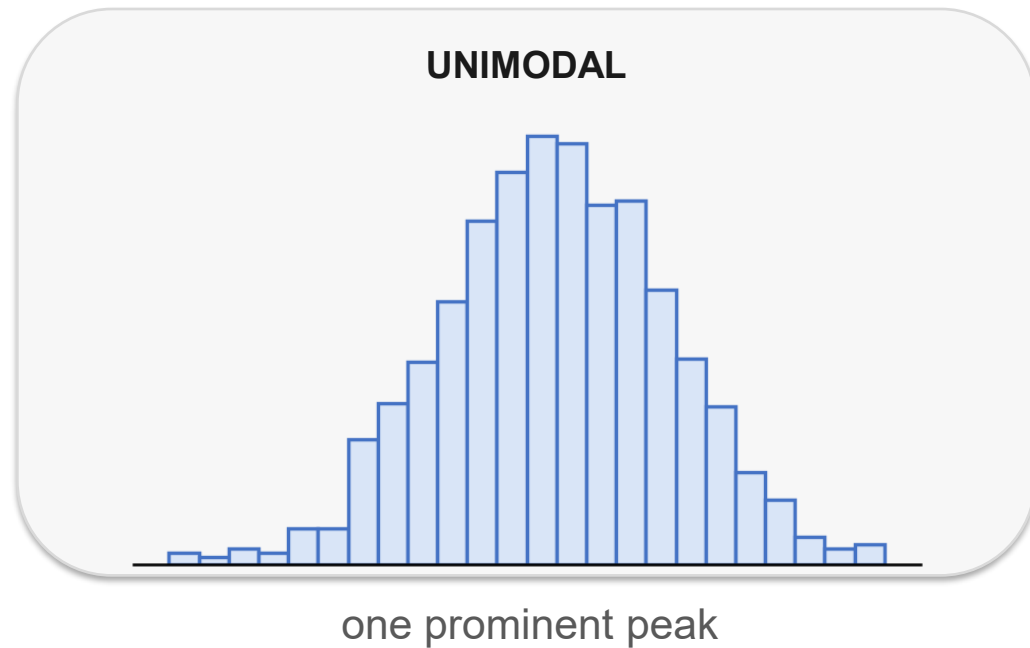
Is the distribution
left-skewed,
symmetric, or
right-skewed?

Answer: left-skewed

Reason: the long tail points toward the lower grades.

Modality: one peak or more than one?

The number of prominent peaks can reveal distinct operating regimes or subgroups.



Bimodality often suggests a mixture—for example, two machines, materials, or process states.

Stem-and-leaf plots: compact display with exact values

Split each observation into a stem and a leaf.

EXAMPLE VALUE

7 | **2**
stem leaf

WHY USE IT?

- Retains every sample value
- Shows concentration, gaps, and shape
- Works best for small or moderate samples
- Requires a key that defines place value

Key: 7 | 2 represents 72 minutes.

Example 7: Old Faithful dormant periods

Duration of dormant periods of the geyser Old Faithful (minutes)

Stem	Leaves
4	2 5 9
5	0 1 1 1 1 3 3 5 5 6 6 7 8
6	0 6 7 7 8 9
7	0 1 2 3 3 4 5 5 5 5 6 6 6 6 6 9 9
8	0 0 0 0 1 2 2 2 3 3 4 4 4 5 6 6 6 8
9	0 1 3

READ THE FIRST ROW

4 | 2 5 9

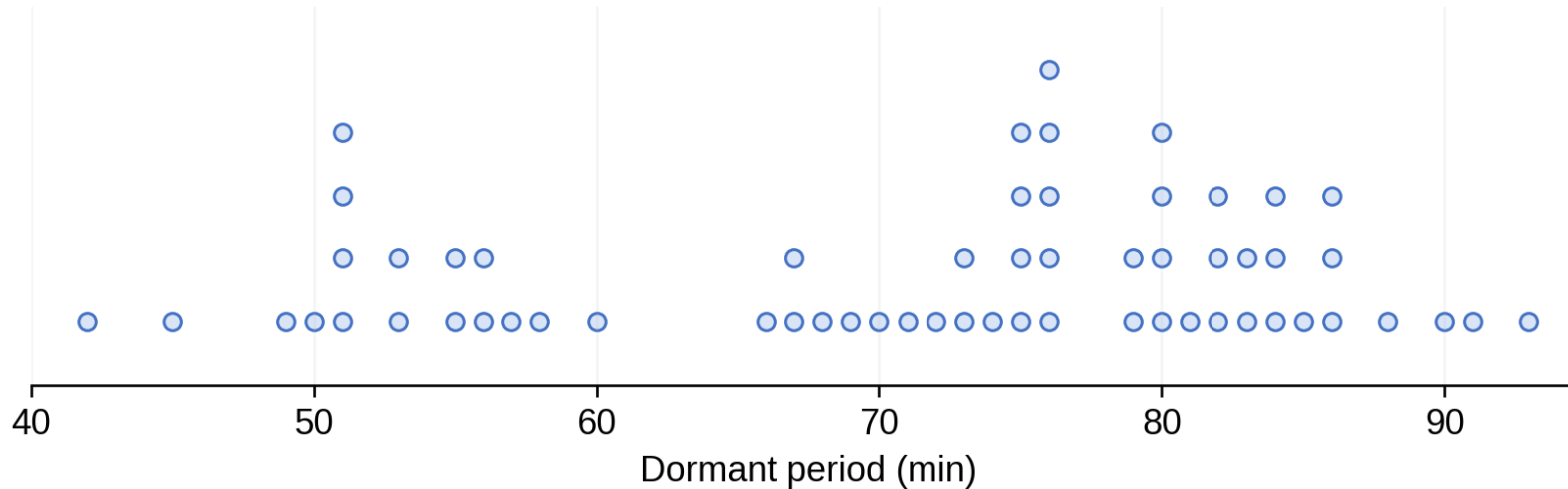
42, 45, and 49 min

All 60 observations can be reconstructed.

What features are visible? Two broad concentrations appear around the 50s and the 70s–80s.

Dotplots: where observations concentrate and repeat

A dotplot places one dot for each observation and stacks repeated values.



BEST FOR

- Small to moderate n
- Repeated values
- Gaps and clusters
- Exact locations

Unlike a stem-and-leaf plot, a dotplot emphasizes location visually rather than listing digits.

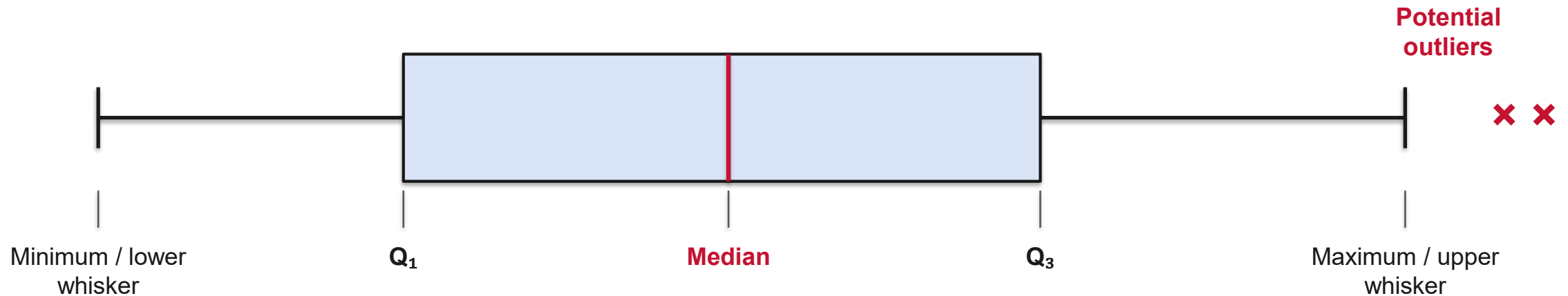
Choosing among a stem plot, dotplot, histogram, and boxplot

DISPLAY	BEST FEATURE	LIMITATION
Stem-and-leaf	Retains exact values and shows shape	Awkward for large samples
Dotplot	Shows clusters, gaps, and repeats	Crowded when n is large
Histogram	Shows distribution shape efficiently	Exact values are lost; bin choice matters
Boxplot	Compact comparison of centre, spread, and outliers	Shape and modality are largely hidden

There is no universally “best” graph; choose the display that answers the question.

Boxplots: a compact five-number summary

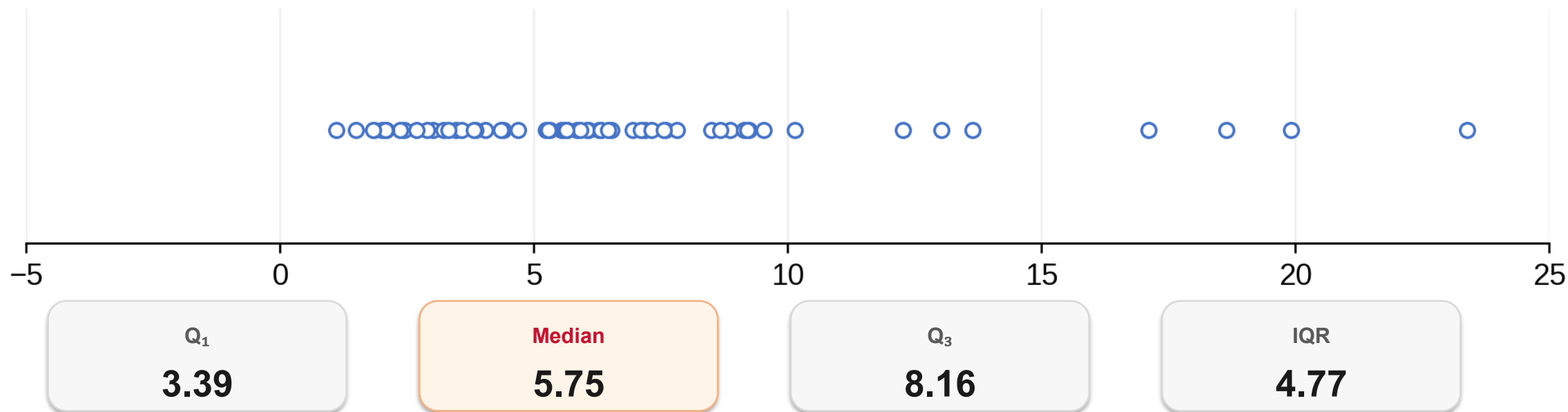
A boxplot displays quartiles, whiskers, and potential outliers on one scale.



Whiskers extend to the most extreme observations inside the $1.5 \times \text{IQR}$ fences.

High-altitude PM emissions: inputs for a boxplot

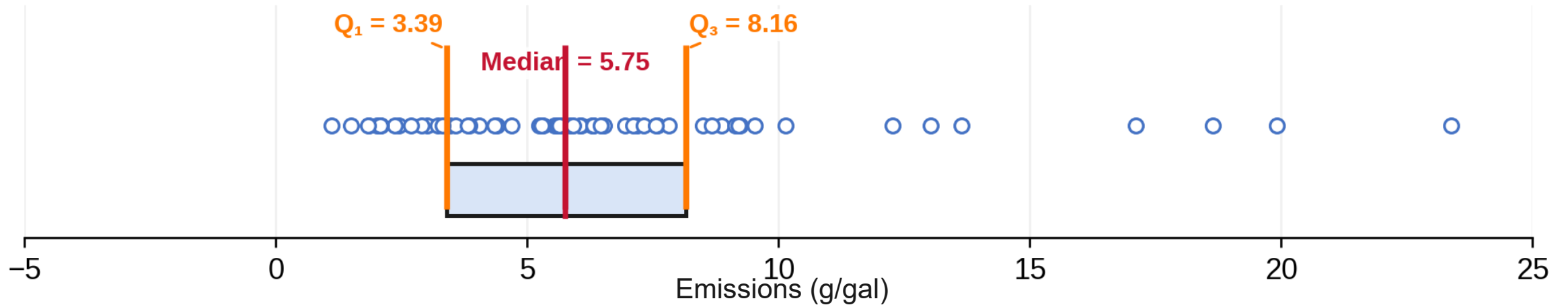
Sample size $n = 62$; emissions are measured in g/gal.



Course percentile rule: use position $p(n + 1)/100$; when the position is not an integer, average the adjacent observations.

Boxplot construction — Step 1: draw the box

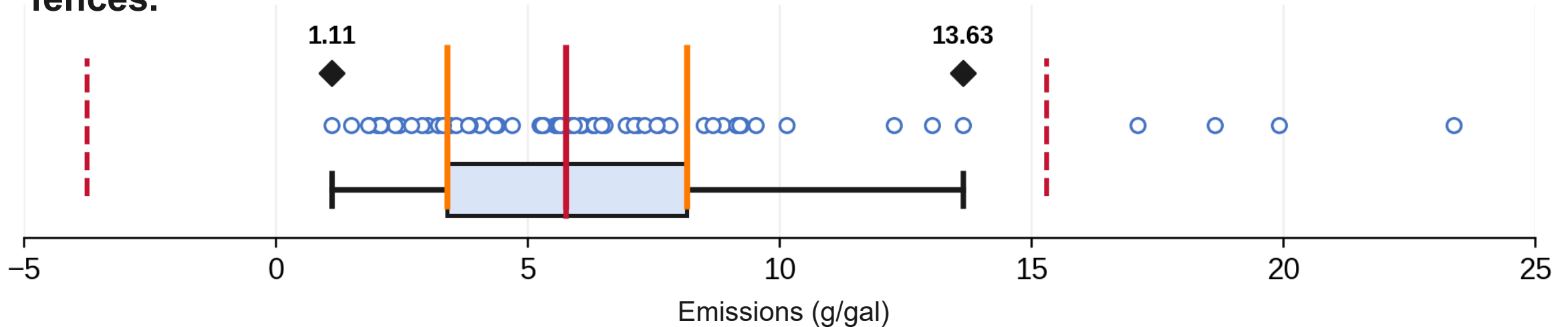
Plot Q_1 , the median, and Q_3 ; then draw the box from Q_1 to Q_3 .



The box spans the middle 50% of the sample; box width on the scale = IQR.

Boxplot construction — Step 2: fences and whiskers

Fences flag potential outliers; whiskers stop at actual observations inside the fences.



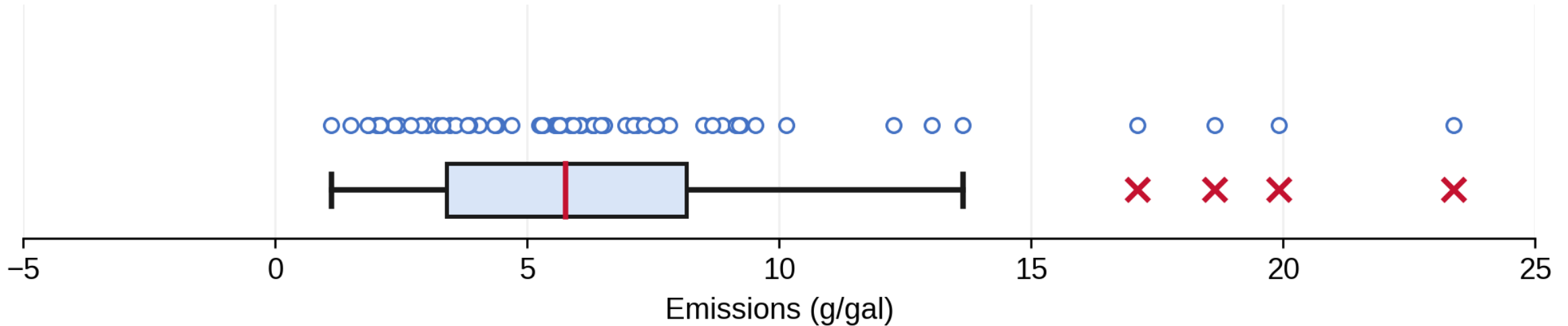
$$\text{Lower fence} = Q_1 - 1.5(\text{IQR}) = -3.758$$

$$\text{Upper fence} = Q_3 + 1.5(\text{IQR}) = 15.303$$

Whiskers end at 1.11 and 13.63—the most extreme observed values inside the fences.

Boxplot construction — Step 3: plot outliers

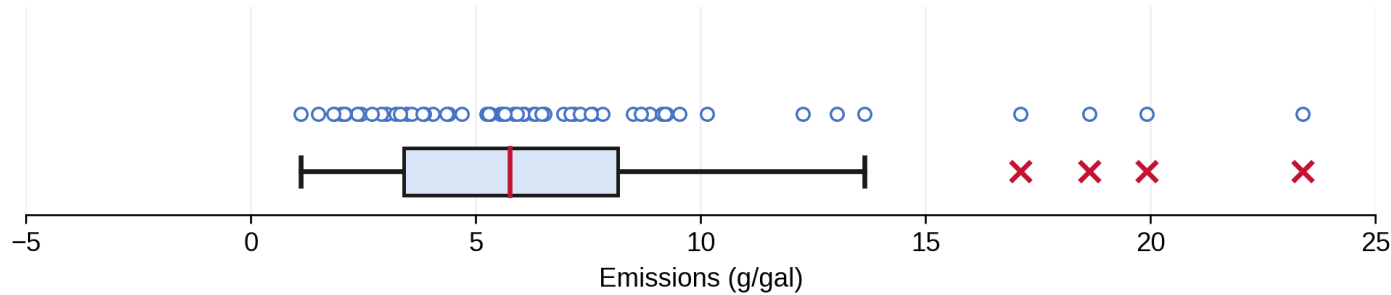
Observations outside the fences are flagged as potential outliers.



Upper outliers: 17.11, 18.64, 19.91, and 23.38 g/gal

“Outlier” is a statistical flag, not a conclusion that the measurement is erroneous.

Interpreting a boxplot

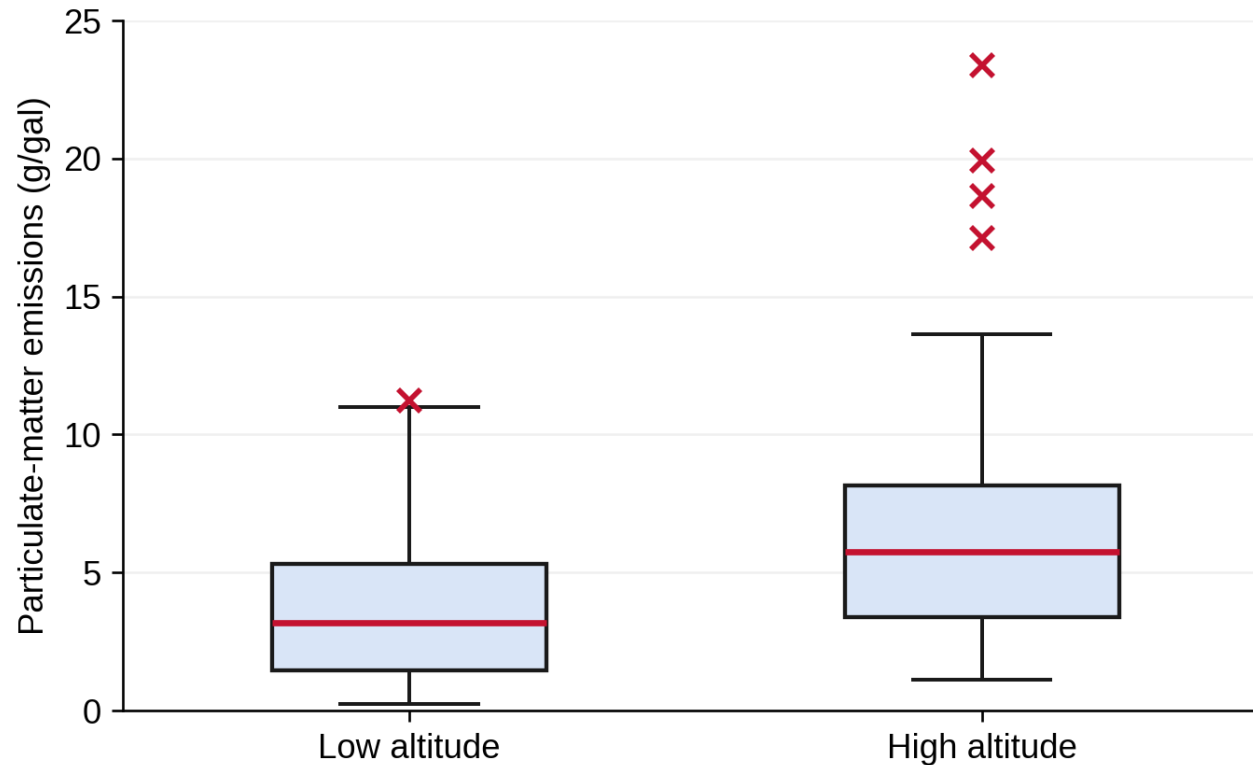


READ IN THIS ORDER

- 1 Centre** Median = 5.75
- 2 Middle spread** IQR = 4.77
- 3 Asymmetry** Longer upper side suggests right skew
- 4 Unusual values** Four high outliers

A boxplot can suggest skewness, but it cannot reveal gaps or multiple peaks.

Comparative boxplots: low vs. high altitude



COMPARE THE SAMPLES

- Centre** High-altitude median is larger.
- Spread** High-altitude IQR is wider.
- Outliers** Both have high outliers; high altitude has more.
- Overall** High-altitude emissions tend to be higher and more variable.

Side-by-side boxplots make group comparison easier than separate raw-data tables.

Comparative boxplots across time of day

Source coefficients were compared at five sampling times.

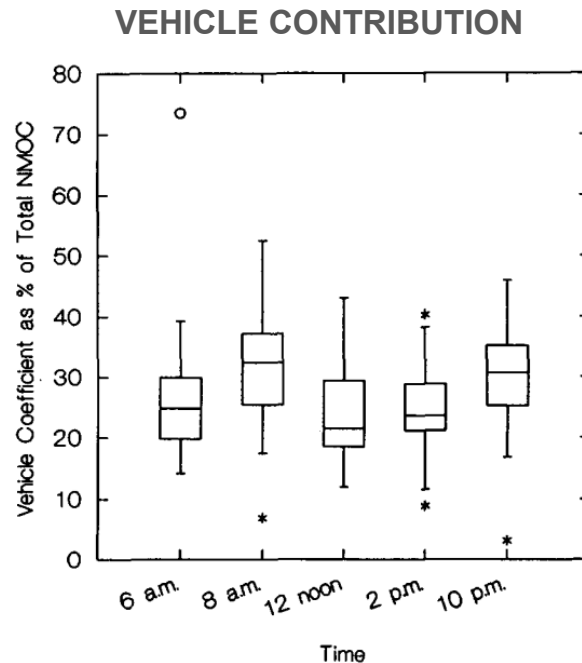


FIG. 1. Distribution of Detroit Vehicle Coefficients

Original interpretation: vehicle contribution peaks near 8 a.m.

GASOLINE-VAPOUR CONTRIBUTION

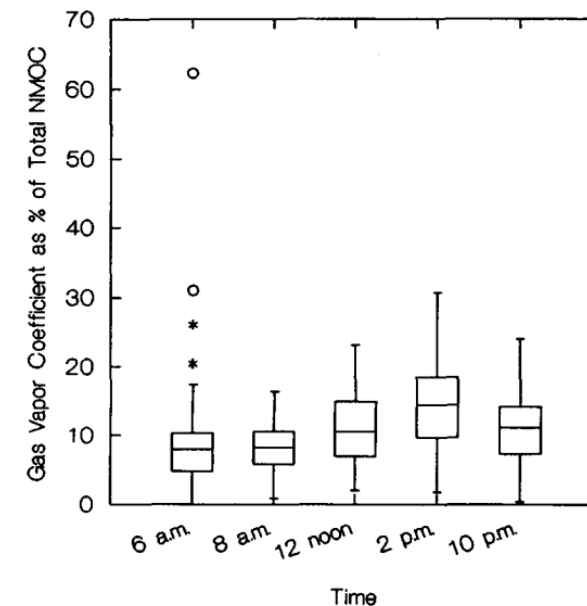


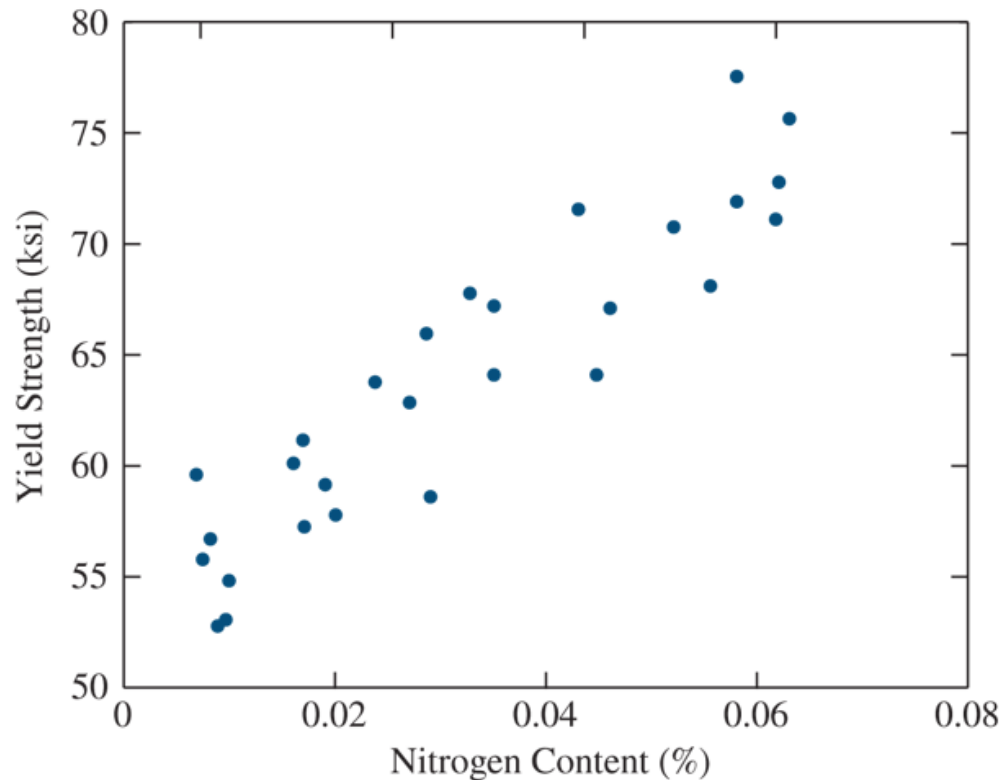
FIG. 2. Distribution of Detroit Gasoline-Vapor Coefficients

Original interpretation: gasoline-vapour contribution peaks near 2 p.m.

Source retained from the 2024 notes: Journal of Environmental Engineering (1995), pp. 483–490.

Scatterplots: graphical summaries for paired data

Bivariate data pair two quantitative measurements for each observational unit.



EXAMPLE

x: nitrogen content (%)

y: yield strength (ksi)

Each dot represents one paired observation.

The upward pattern indicates a positive association.

Looking at scatterplots

Describe a scatterplot using four features.

DIRECTION

Positive, negative, or no apparent direction

FORM

Linear, curved, clustered, or another pattern

STRENGTH

How tightly points follow the form

UNUSUAL POINTS Outliers or high-leverage observations

Association can support prediction, but it does not establish causation.

Key takeaways

- Match the display to the variable type, sample size, and question.
- Frequency tables group quantitative observations into class intervals.
- Histogram shape is described by centre, spread, skewness, modality, gaps, and outliers.
- With unequal class widths, histogram area represents relative frequency.
- Stem-and-leaf plots and dotplots retain detailed information for smaller samples.
- Boxplots compactly display quartiles, whiskers, and potential outliers and are excellent for comparison.
- Scatterplots reveal relationships between paired quantitative variables.

Next: probability models for quantifying uncertainty.